

Kevin Sann

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MISSION STATEMENT

I am a research scientist solving problems for national security at the intersection of hardware, software, and computer security. My experiences span all three across the semiconductor value chain, government contracting, and academia in my ownership of products and prototypes. Ideation and execution based upon rigorous ground-truth findings reinforce my successes in these arenas. My primary goal is to advance adoption of performant, secure computer architectures and explore versatile ways to rewrite the hardware-software contract by extending it or changing its preexisting clauses. So far, I have paid tribute to this goal with my academic research scrutinizing and modifying an existing secure architecture codesign to maximally reuse hardware. Additional experiences supporting my goal come from system software and compilers research for two discrete secure architectures.

EDUCATION

UNIVERSITY OF MINNESOTA, Minneapolis, MN
Spring 2024 Graduate Non-Degree Coursework
F. 2020-F. 2023 Bachelor of Computer Engineering

RESEARCH EXPERIENCE

2024-present **Research Scientist – Systems/Compilers**, Riverside Research Institute, Lexington, MA
Secure & Resilient Systems group; research in a non-profit setting for national security.

Rust on Tagged Architectures (ROTA)

- Ported the rust compiler to a secure processor architecture and its ecosystem
- Built LLVM backend from scratch referencing industry-strength targets and GCC backend
- Designed a custom LLVM pass to lower programmer interfaces to hardware features

Custom ASIC Follow-on

- Awarded a follow-on contract to ROTA incorporating formal methods in the ecosystem
- Produced findings of the secure processor microarchitecture to examine gate-level security guarantees
- Demonstrated HDL-level compile-time information flow security so insecure designs never compile

CHERI Fuzz

- Enabled the first successful pure-capability CHERIBSD Morello port of AFL++ instrumentation
- Designed and built executor and feedback modules for blackbox fuzzing in Draper Laboratory's VMF
- Authored surveys on state-of-the-art to normalize measurement, fitness metrics for blackbox fuzzing

DARPA E-BOSS

- Ported AFLGo from Ubuntu 20.04 to Ubuntu 22.04 for dynamic analysis feedback to static analysis tooling during construction of an enhanced software bill-of-materials (SBOM)

Binary Program Analysis Survey

- Authored a survey on the state-of-the-art for static and dynamic binary rewriting approaches

F. 2023-Spr. 2024 **Academic Research**, University of Minnesota, Minneapolis, MN

Minimal Minos

Minimal Minos focuses on minimizing microarchitecture changes in Minos, a co-design enforcing Control Flow Integrity whose call for extra DRAM bus width and a separate chip have been criticized as invasive.

- Developed a co-design replacing extra off-chip hardware by extending TLB entries with pointers to integrity pages and stuffing an access to integrity information of each entry right after any physical access.
- Evaluated Bochs' quality of emulation cross referencing the original Netburst microarchitecture and true Out-Of-Order (OOO) execution, finding several discrepancies.
- Full paper: [\[link\]](#)

PROFESSIONAL EXPERIENCE

Summer 2024 **Compiler Engineer**, Texas Instruments, Sugar Land, TX
Authored code for the proprietary, in-house TI C7000 VLIW Digital Signal Processor C/C++ compiler toolchain.

- Removed duplicate instruction sequences with pass logic based on SSA form, dead code elimination
- Replaced cc1 macro string concatenation accessors with parser operators based on OpenCL operators

Summer 2023 **Product Engineer**, Texas Instruments, Tucson, AZ
Owned products at ATP in the semiconductor value chain taking a development-stage family to production.

- Allowed swapping commercial die for excess automotive die, releasing 50 million backlogged parts
- Received the ETS-364 Basic Analog, Digital, & Pattern Based Testing certification.

Summer 2022 **Product Engineer**, Texas Instruments, Dallas, TX
Owned products at ATP in the semiconductor value chain taking spins on an established product to production.

- Qualified a second source (production flow) for power stage devices, supporting 20 million parts
- Enforced quality & reliability requirements based on datasheet parameters
- Facilitated engineering prototypes through ATE readpoint testing and reliability/stress testing